Data Analysis with NumPy and Pandas

Introduction to Data Analysis with Python - Week 7

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What is NumPy?

- Numerical Python: the foundation of the scientific Python stack
- Core object: the ndarray, an N-dimensional array
- Fast: operations run in compiled C code
- Pandas, SciPy and scikit-learn build on it

Why not just use Python lists?

Python list

- Can hold mixed types
- Slower, uses more memory
- Needs explicit loops

```
[x * 2 for x in [1, 2, 3]]
```

NumPy array

- One data type, contiguous memory
- Fast, compiled operations
- Vectorized: no loops needed

```
np.array([1, 2, 3]) * 2
```

NumPy features

ndarray

Fast, homogeneous, N-dimensional

Vectorization

Replaces loops

Boolean masks

Filter data with conditions

axis

Aggregate by rows or columns

Broadcasting

Handles mismatched shapes

Next: Pandas

Builds on all of this

What is Pandas?

- Library for tabular, labeled data
- Like Excel or SQL, inside Python
- Built on top of NumPy
- Handles missing values, mixed types and dates
- Main objects:
    - **Series**: a labeled 1d array
    - **DataFrame**: a 2D table, where each column is a series

Pandas features

Series, DataFrame

The core structures

Inspect first

head, info, describe

Select and filter

loc, iloc, boolean conditions

Missing values

Handle them explicitly

groupby and merge

Your most powerful tools

Next: workflow

Put it all together

The data analysis workflow

1. **Ask a question**

   What do you need to know?

2. **Load the data**

   read_csv, read_excel

3. **Inspect and clean**

   info, isna, dtypes

4. **Explore, transform**

   Filter, new columns

5. **Analyze**

   groupby, aggregations

6. **Communicate**

   Charts and conclusions